

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS

DATE: NOVEMBER 21st, 2022

PERMITTED AIDS: Approved simple calculator

THE EXAM CONSISTS OF 5 PROBLEMS ON 5 PAGES, 9 PAGES OF ENCLOSURES.

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All points a), b), c) and so on are weighted equally. There are a total of 21 such points.

Problem 1

Consider a random variable X with probability density

$$f(x) = \frac{1}{4C} \exp\left(\frac{x}{4}\right), \quad 0 \leq x \leq 4.$$

Further, it can be shown that $\text{Var}(X) = 16(1 - C^{-1} - C^{-2})$.

- Show that $C = \exp(1) - 1 \approx 1.718281828$
- Show that $E(X) = 4/C$ and write down the integral needed to find $\text{Var}(X)$.
- Show that the cumulative distribution function $F(x)$ is given by $F(x) = \frac{\exp(x/4)-1}{C}$, $0 \leq x \leq 4$.
Use $F(x)$ to calculate $P(X > 1)$ and $P(1 < X < 2)$
- Derive the inverse cumulative distribution function and write down the algorithm (with words and formulas, not code) required to simulate n random variables with density $f(x)$ using the inverse transform method.

Consider the following R-code:

```

1  # The pdf of problem 1
2  p1.pdf <- function(x){return(0.25/(exp(1)-1.0)*exp(0.25*x))}
3  # First proposed AR-method:
4  p1.ar.1 <-function(){
5    c <- 4.0*p1.pdf(4.0)
6    while(TRUE){
7      x <- runif(1,min=0,max=4)
8      if(runif(1) < p1.pdf(x)/(c*dunif(x,min=0,max=4))){return(x)}
9    }
10 }
11 # Second proposed AR-method:
12 p1.ar.2 <-function(){
13   c <- 4.0*p1.pdf(4.0)
14   while(TRUE){
15     x <- 4.0*runif(1)
16     if(runif(1) < p1.pdf(x)/c){return(x)}
17   }
18 }
```

- Do both of the two above R-functions `p1.ar.1()` and `p1.ar.2()` return correct samples with density $f(x)$ using the acceptance-rejection method with $Uniform(0, 4)$ proposal distribution? Justify your answer.

Now consider $n > 30$ iid random variables $X_1, \dots, X_n$, each having the density $f(x)$ above.

- f) Which distribution does $\bar{X} = n^{-1} \sum_{i=1}^n X_i$ approximately have? Please provide the reasoning leading to your answer.

Suppose now you wish to estimate $\theta = E(\sqrt{X})$ based on n iid samples with density $f(x)$ stored in a vector $\mathbf{x}$. Consider the following R-code:

```

1 point.1 <- mean(sqrt(x))
2 point.2 <- sqrt(mean(x))
3 point.3 <- mean(sqrt(x>0))
4 CI.1 <- c(point.1 - 1.96*sd(x)/sqrt(n), point.1 + 1.96*sd(x)/sqrt(n))
5 CI.2 <- c(point.1 - 1.96*sd(sqrt(x))/sqrt(n), point.1 + 1.96*sd(sqrt(x))/sqrt(n))
6 CI.3 <- c(point.2 - 1.96*sd(x), point.2 + 1.96*sd(x))
7 CI.4 <- c(point.2 - 1.96*sqrt(sd(x))/sqrt(n), point.2 + 1.96*sqrt(sd(x))/sqrt(n))
8 CI.5 <- c(point.3 - 1.96*sd(sqrt(x)), point.3 + 1.96*sd(sqrt(x)))
9 CI.6 <- c(point.3 - 1.96*sd(sqrt(x))/n, point.3 + 1.96*sd(sqrt(x))/n)

```

- g) Which of point.1, point.2, point.3 is an unbiased estimate of θ ? Explain.
Which of CI.1, ..., CI.6 is an approximate 95% confidence interval for θ .

Problem 2

Consider a homogeneous Poisson process $N_1(t)$, $t \geq 0$ with intensity λ , and let T_1 be the first event time.

- a) Which distribution does T_1 have? Write down the probability density function of T_1 . Also find $E(T_1^2)$.
b) Find the probability no events had occurred in time 0.1, given that $N_1(1) = 8$, namely $\theta = P(N_1(0.1) = 0 | N_1(1) = 8)$. You can solve this exercise either by theoretical calculations or by describing how you would simulate such a quantity.

Now consider the non-homogeneous Poisson process $N_2(t)$, $t \geq 0$ with intensity $\lambda(t) = 1 + t/10$.

- c) Which distribution does $N_2(5)$ have?

Problem 3

Consider the following integrals:

$$A = \int_0^\infty \exp(-2x)x^{3/2}dx$$

$$B = \int_0^2 \exp(-2x)x^{3/2}dx$$

$$C = \int_{-1}^1 \sqrt{1+x^3}dx$$

$$D = \int_{-1}^1 \sqrt{1+x^4}dx$$

For reference, the values of the integrals are $A \approx 0.2349964007$, $B \approx 0.1982815906$, $C \approx 1.952757241$ and $D \approx 2.178858826$. Further, we have that $\Gamma(3/2) = \sqrt{\pi}/2$ and $\Gamma(5/2) = 3\sqrt{\pi}/4$.

- Write down an expression for a Monte Carlo estimator of A based on n random draws from an $Exponential(1/2)$ distribution (i.e. with mean $\beta = 1/2$).
- Write down an expression for a Monte Carlo estimator of B based on n random draws from an $Exponential(1/2)$ distribution (i.e. with mean $\beta = 1/2$).
- Write down expressions for Monte Carlo estimators of C and D based on n random draws from a $Uniform(-1, 1)$ distribution.
- Write down expressions for importance sampling estimators of A and B based on n random draws from a $Gamma(3/2, 1/2)$ distribution (i.e. with shape parameter α equal to $3/2$ and scale parameter β equal to $1/2$). Do you, based on theory, expect these to work well?
- Write down expressions for importance sampling estimators of A and B based on n random draws from a $Gamma(5/2, 1/2)$ distribution (i.e. with shape parameter α equal to $5/2$ and scale parameter β equal to $1/2$). Do you, based on theory, expect these to work well?
- Write down expressions for antithetic Monte Carlo estimators of C and D based on random draws from a $Uniform(-1, 1)$ distribution and n evaluations of the integrands. Do you, based on theory, expect these to work well?

Problem 4

In mandatory assignment 3, problem 3, bootstrapping of an estimator for $f(1)$ was considered. Here $f(x)$ was an unknown density, and we had a data set with $n = 239$ observations that were distributed iid according to $f(x)$. Below is some code that loads the data set, defines the estimator of $f(1)$ and produces $B = 2000$ bootstrap samples of the estimator stored in a vector `f1.hat.boot`:

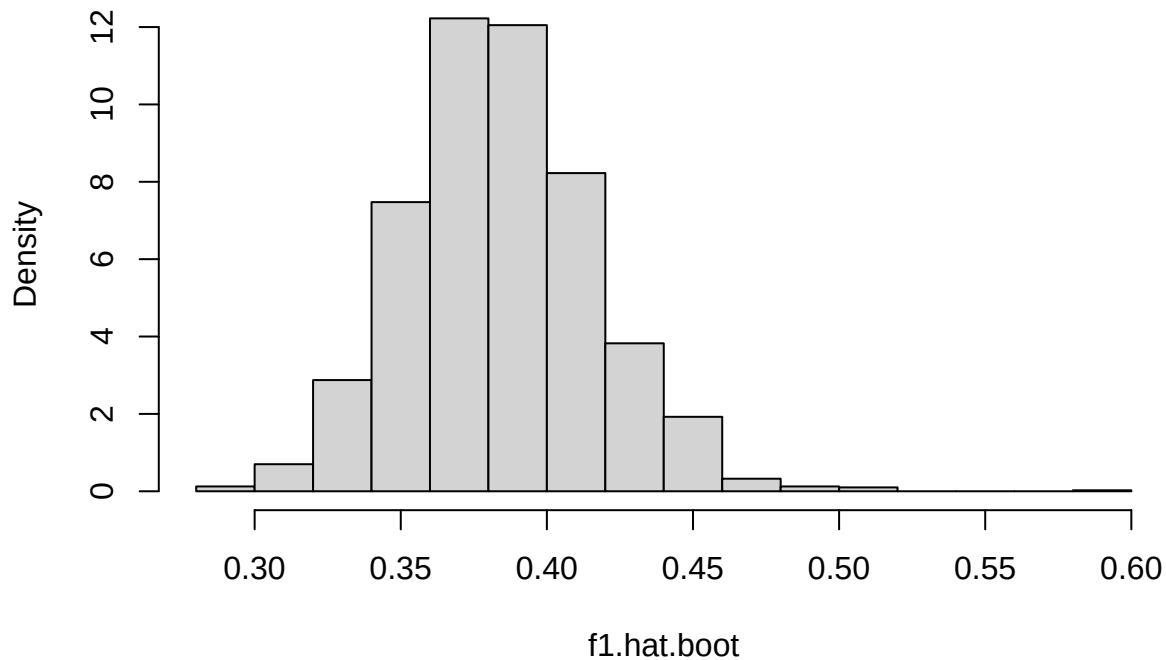
```
x <- read.table("M3data.txt")$V1 # read the data
# estimator of f(1) based on KDE:
f1.hat <- function(x){return(density(x,from=1,to=1,n=1)$y)}
# estimate of the full data set:
f1.o <- f1.hat(x)
print(paste0("Estimate of f(1) based on data set : ",f1.o))

## [1] "Estimate of f(1) based on data set : 0.383732864871009"

# now do the bootstrapping
B <- 2000
f1.hat.boot <- replicate(B,f1.hat(x[sample(1:length(x),
                                         size=length(x),replace=TRUE)]))

# display the bootstrap samples
hist(f1.hat.boot,probability = TRUE)
```

Histogram of f1.hat.boot



Below is a set of summary statistics of `f1.hat.boot`

```
print(mean(f1.hat.boot))
```

```
## [1] 0.3841279
```

```
print(median(f1.hat.boot))
```

```
## [1] 0.382843
```

```
print(var(f1.hat.boot))
```

```
## [1] 0.001026572
```

```
print(round(quantile(f1.hat.boot, probs=c(0.01,0.025,0.05,0.1,0.9,0.95,0.975,0.99)),3))
```

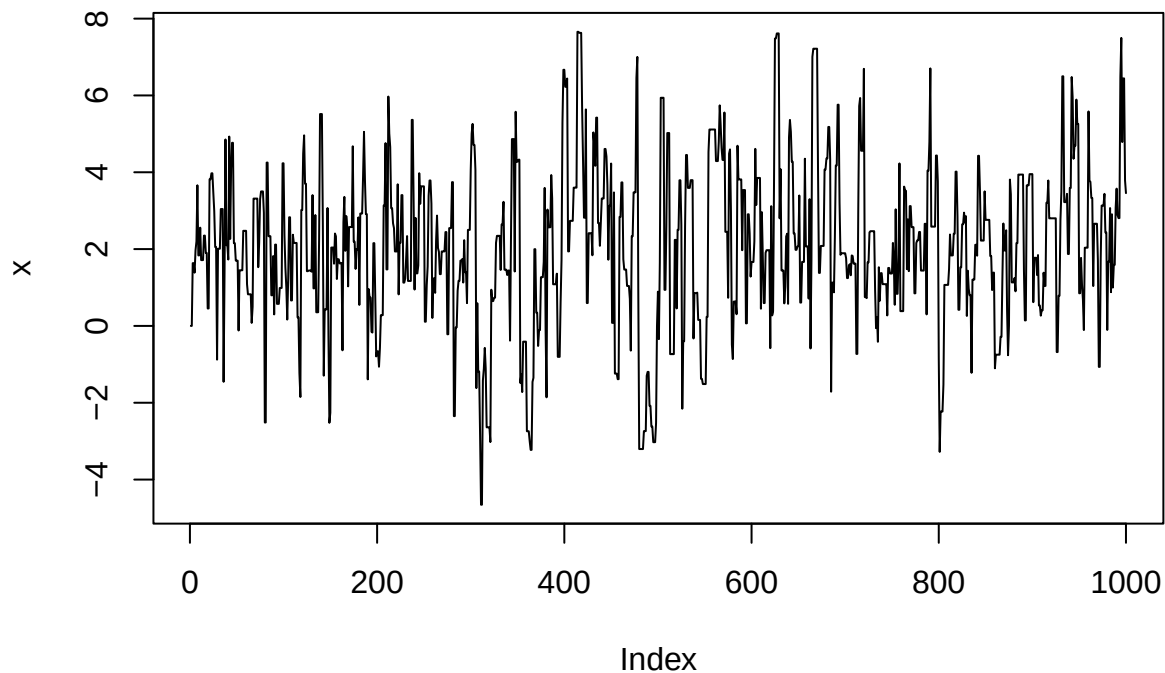
```
##      1%  2.5%   5%   10%   90%   95%  97.5%  99%  
## 0.314 0.327 0.335 0.345 0.425 0.440 0.449 0.463
```

- Find estimates of the bias, standard error and MSE of the estimator.
- Find a 95% standard normal bootstrap interval for $f(1)$.
- Find a 95% percentile bootstrap interval for $f(1)$. Comment on any differences (or lack thereof) between the intervals in b) and c).

Problem 5

Below is the code of a specific type of MCMC algorithm targeting a 1-dimensional probability distribution. The algorithm is first run, and then a traceplot is produced.

```
n.iter <- 1000
x <- numeric(n.iter)
x[1] <- 0.0
for(t in 2:n.iter){
  x.prop <- x[t-1] + 3*rnorm(1)
  alpha <- min(1.0,dnorm(x.prop,mean=2,sd=2)/dnorm(x[t-1],mean=2,sd=2))
  if(runif(1)<alpha){
    x[t] <- x.prop
  } else {
    x[t] <- x[t-1]
  }
}
plot(x,type="l")
```



- a) Which type of MCMC algorithm is it?
Which proposal distribution is used?
What is the target distribution/stationary distribution of the above MCMC algorithm?

Now suppose we calculate the average of the last 500 iterations, say `mean(x[501:1000])`

- b) Would you expect that this average would have a smaller/equal/larger standard deviation than a hypothetical average calculated from 500 iid samples from the target distribution? Explain.